DSCI 510 Final Project Proposal: ***Project Title: E-Commerce Price Comparison and Trend Analysis Across Multiple Retailers***

Team Members:

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1. Problem Statement

Prices for identical or similar products vary significantly across online retailers, but these differences are not easily visible to consumers. This project aims to analyze and compare product pricing, discount patterns, and rating differences across multiple e-commerce websites. By scraping real product listings and conducting statistical analysis, the goal is to identify which platform offers consistently lower prices and whether higher prices correlate with better customer ratings.

2. Data Collection Plan

Data will be collected through web scraping static HTML product listing pages from retailers such as Walmart, Target, and eBay. Using Python's *requests* and *BeautifulSoup* libraries, we will extract product names, prices, original prices when available, ratings, review counts, availability or shipping indicators, and the platform name.

3. Analysis Plan

Analysis will be conducted using Pandas and NumPy. The project will compare average prices across websites, examine price variation within each retailer, evaluate the relationship between ratings and prices, analyze discount percentages between platforms, and identify the platform offering the lowest overall prices for the chosen product category.

4. Visualization Plan

Visualizations created using Matplotlib will include bar charts comparing average prices, boxplots showing price distributions, scatter plots illustrating rating–price relationships, bar charts comparing discount percentages, and histograms showing price ranges within each retailer. These visualizations will effectively communicate pricing and rating patterns across platforms.

5. Expected Outcome

The final outcome will be a fully cleaned e-commerce dataset, a complete analysis workflow, and visualizations that highlight pricing differences, discount behavior, and rating patterns across major retailers. The project will reveal which platform is most cost-effective and provide meaningful insights into how prices and ratings vary across online marketplaces.